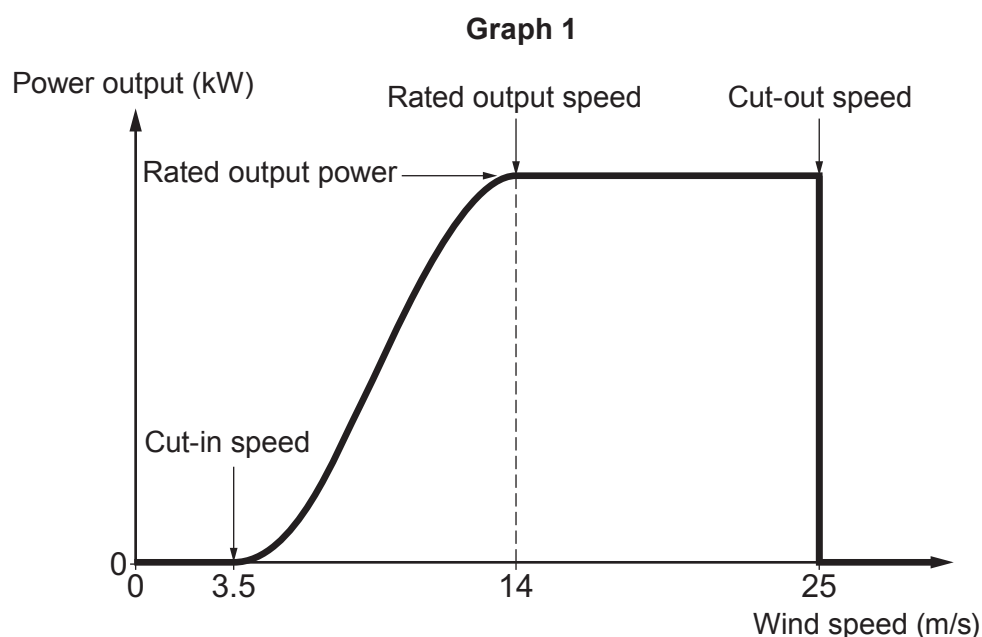


7. Wind turbines are mounted on towers to capture the most energy. They harness the wind's energy with their propeller-like blades. Wind turbines can be used as stand-alone applications, or they can be connected to the National Grid. For larger scale sources of wind energy, a number of wind turbines are built close together to form a wind farm. Several electricity providers today use wind farms to supply power to their customers.

- (a) The power output from a wind turbine is not constant but depends on wind speed as shown in Graph 1.



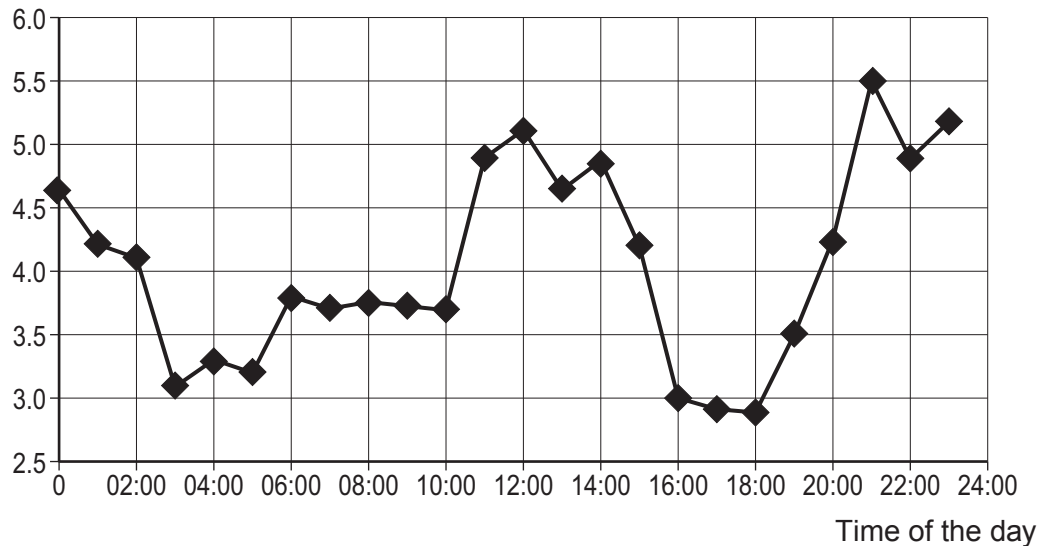
The cut-in speed is the speed at which the turbine first starts to rotate and generate power. Somewhere between 12 and 15 m/s, the output power reaches the maximum limit. This limit is called the rated output power and the wind speed at which it is reached is called the rated output speed. As the speed increases the forces on the turbine structure continue to rise and, at some point, there is a risk of damage to the rotor. As a result, a braking system is employed to bring the rotor to a standstill. This occurs at the cut-out speed and is usually around 25 m/s.



The mean wind speed for different times during one day in May are shown in Graph 2.

Graph 2

Mean wind speed (m/s)



It is claimed that wind power is **always** a useful back up at times of high demand between the hours of 06:00 to 10:00 and between 16:00 to 20:00 because it is always windy.

Explain whether the information in both Graphs 1 and 2 support this claim.

[4]

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TURN OVER FOR THE REST OF THE QUESTION



- (b) A wind farm off the coast of Wales, has an output voltage of 16 kV and a total rated output power of 24 MW. It is connected to the National Grid by a transformer. The transformer has an input coil containing 400 turns and an output coil containing 4 800 turns. Assume the transformer has an efficiency of 100%.

Use equations from page 2 to answer the following questions.

- (i) Calculate the output voltage of the transformer in kV. [2]

Output voltage = kV

- (ii) Calculate the output current of the transformer. [3]

Output current = A

- (c) (i) Explain the role of transformers in the National Grid. [3]

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- (ii) Explain the purpose of an iron core in a transformer **and** state why it is laminated. [3]

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Examiner
only

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END OF PAPER

